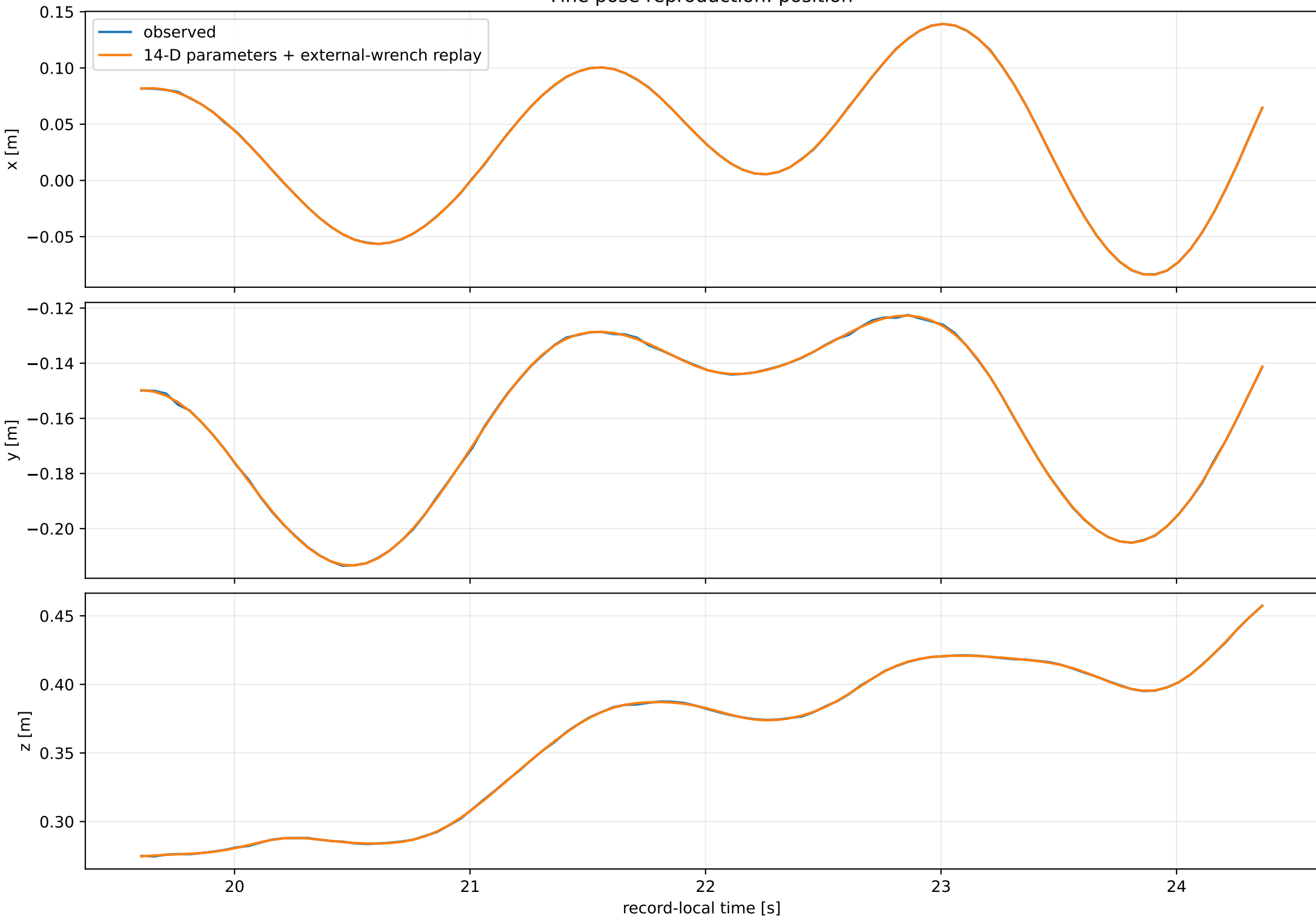
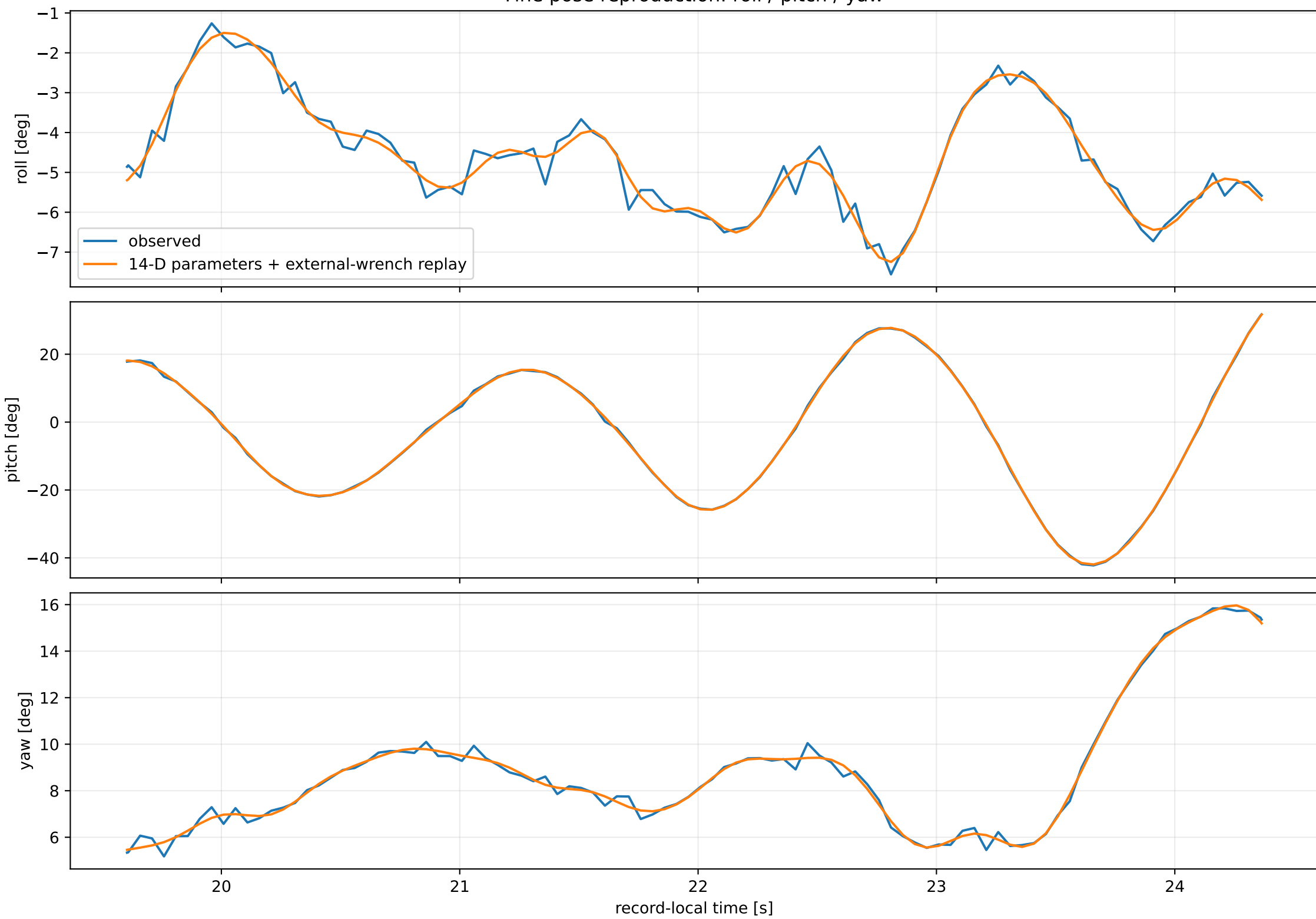


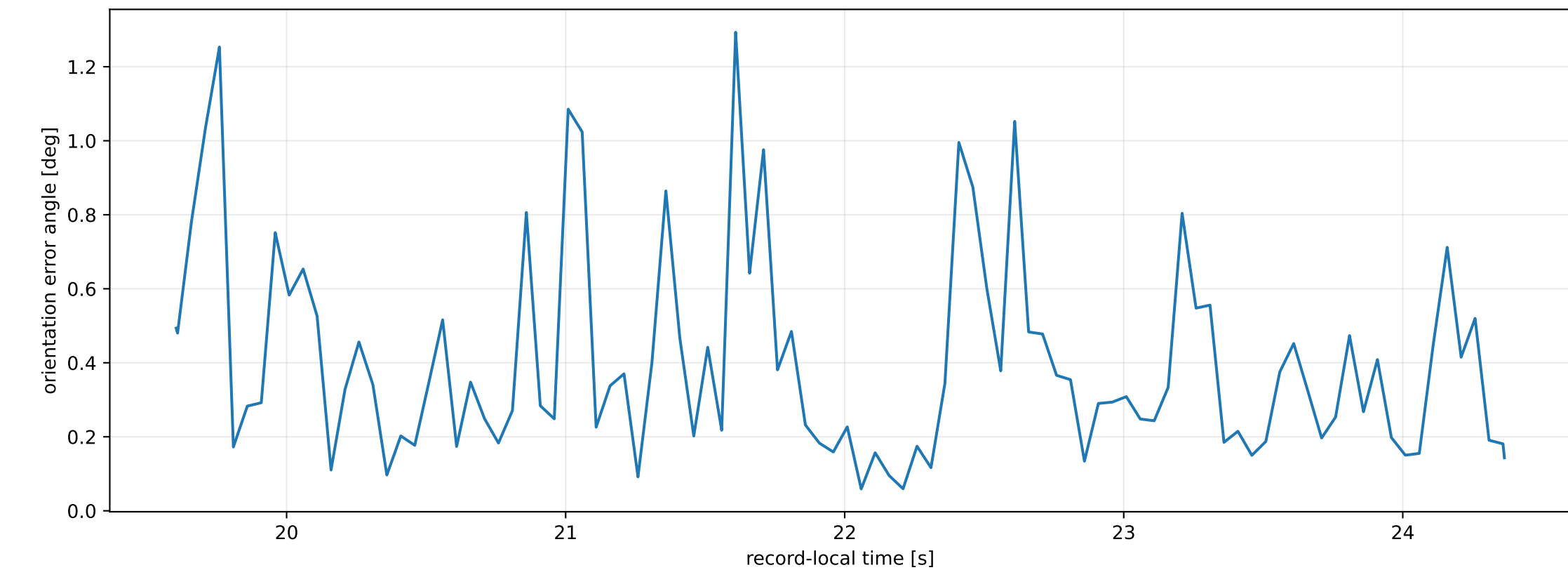
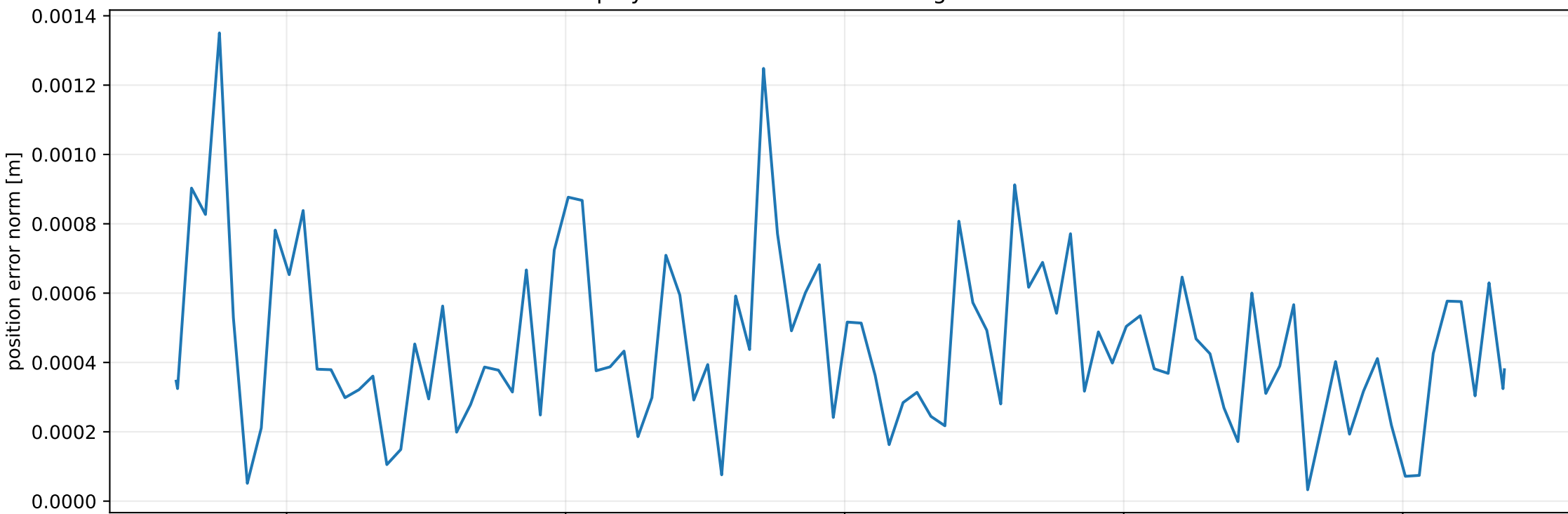
Fine pose reproduction: position



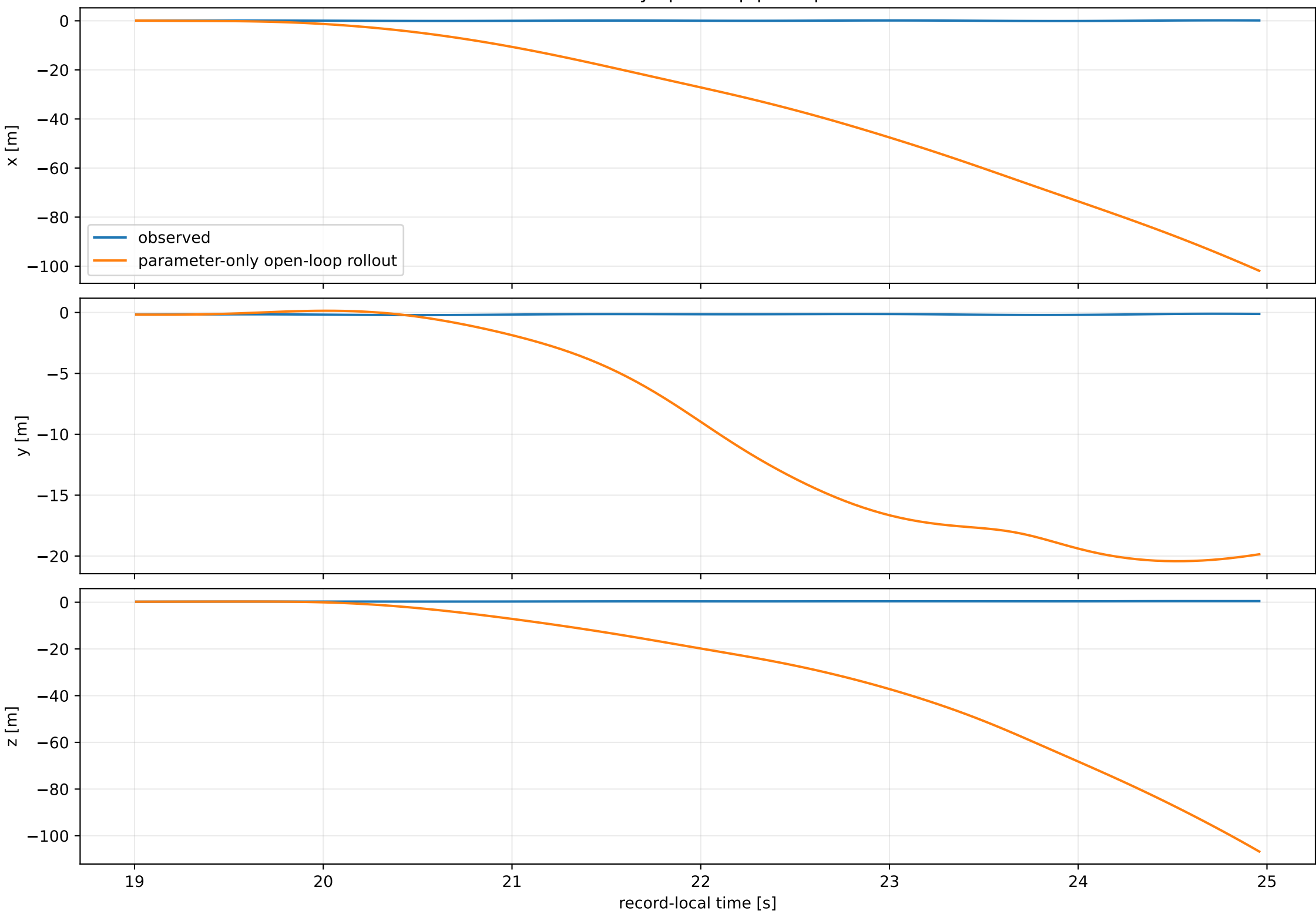
Fine pose reproduction: roll / pitch / yaw



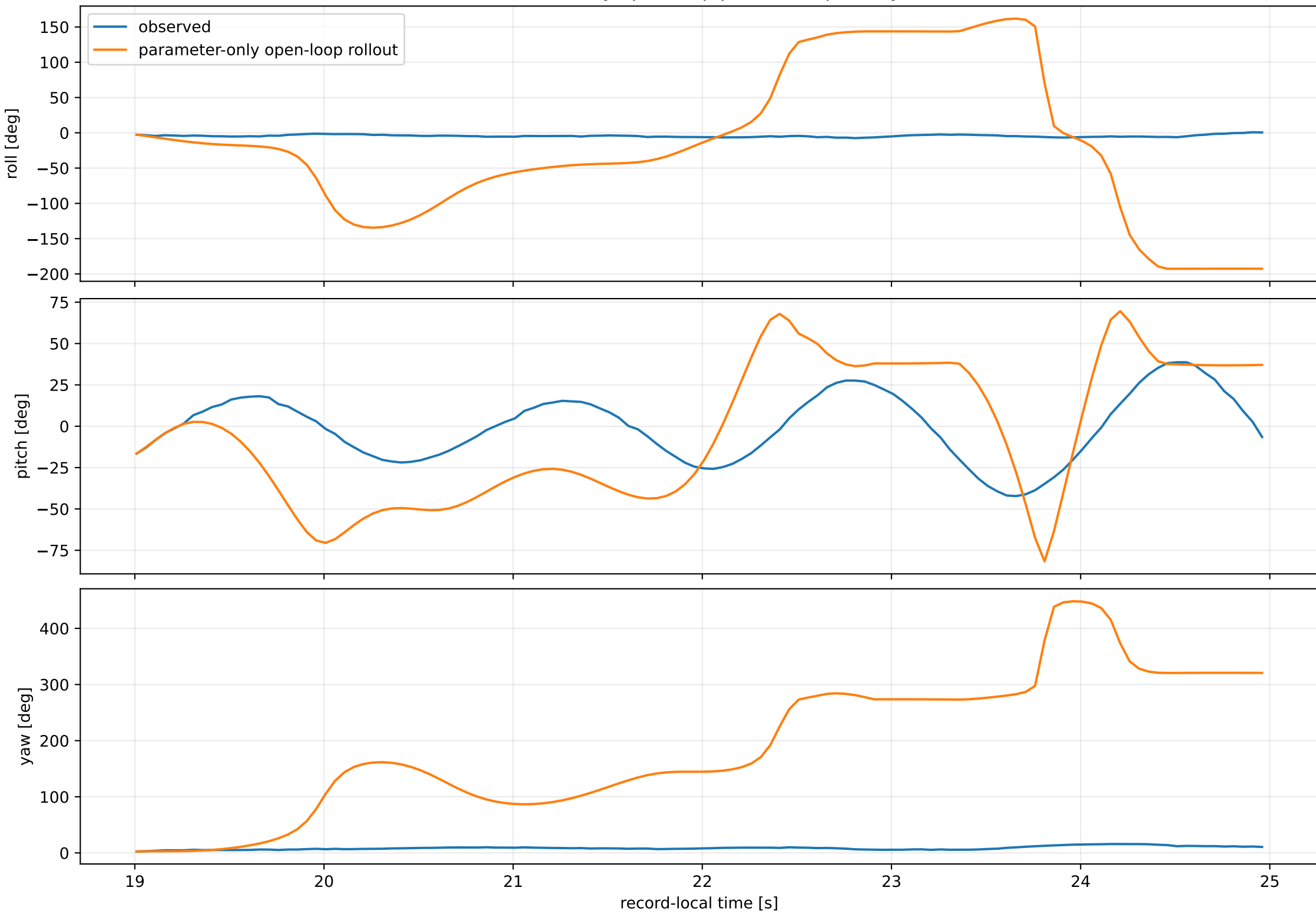
Fine replay error after the 14-D change — lower is better



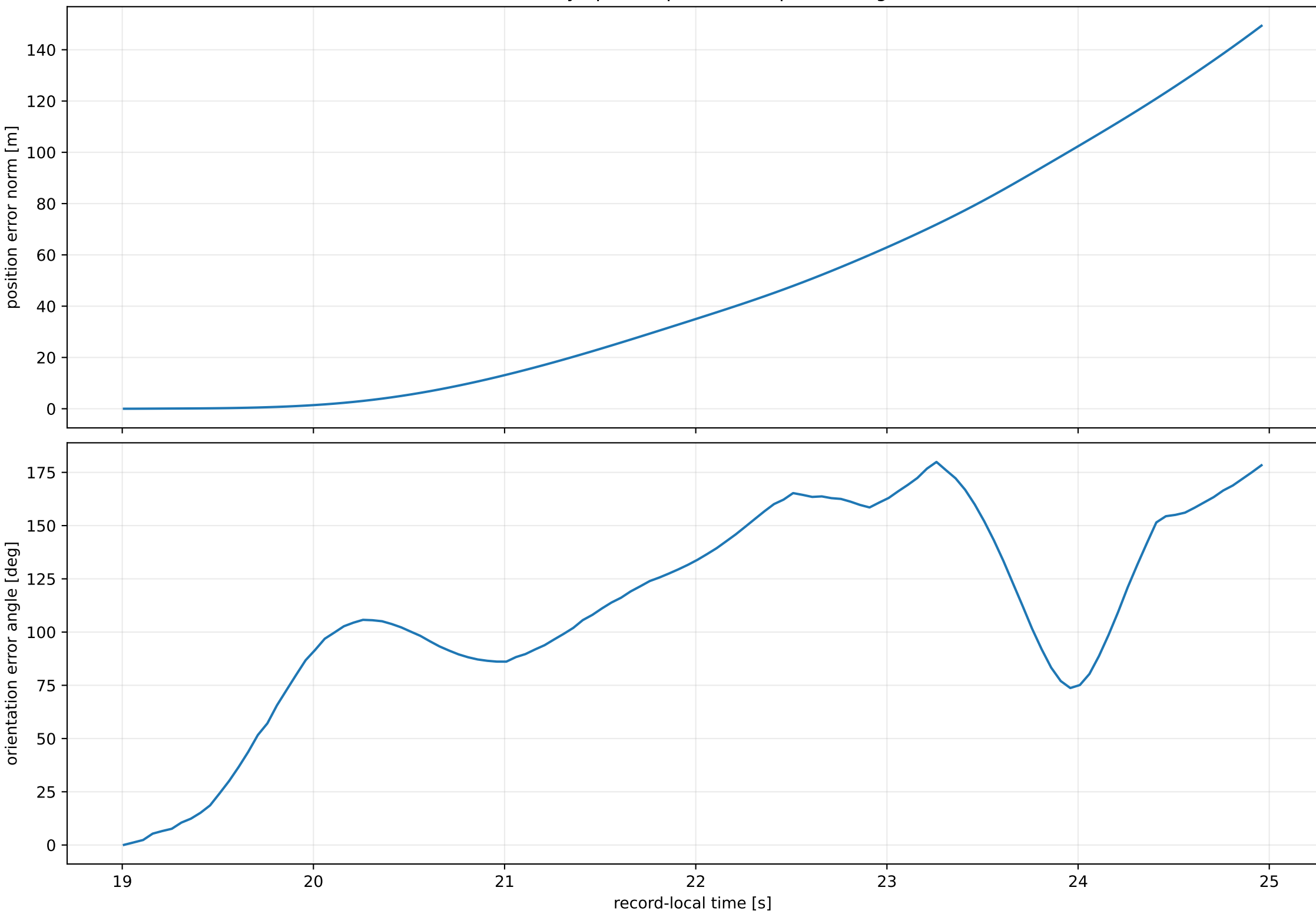
Parameter-only open-loop pose: position



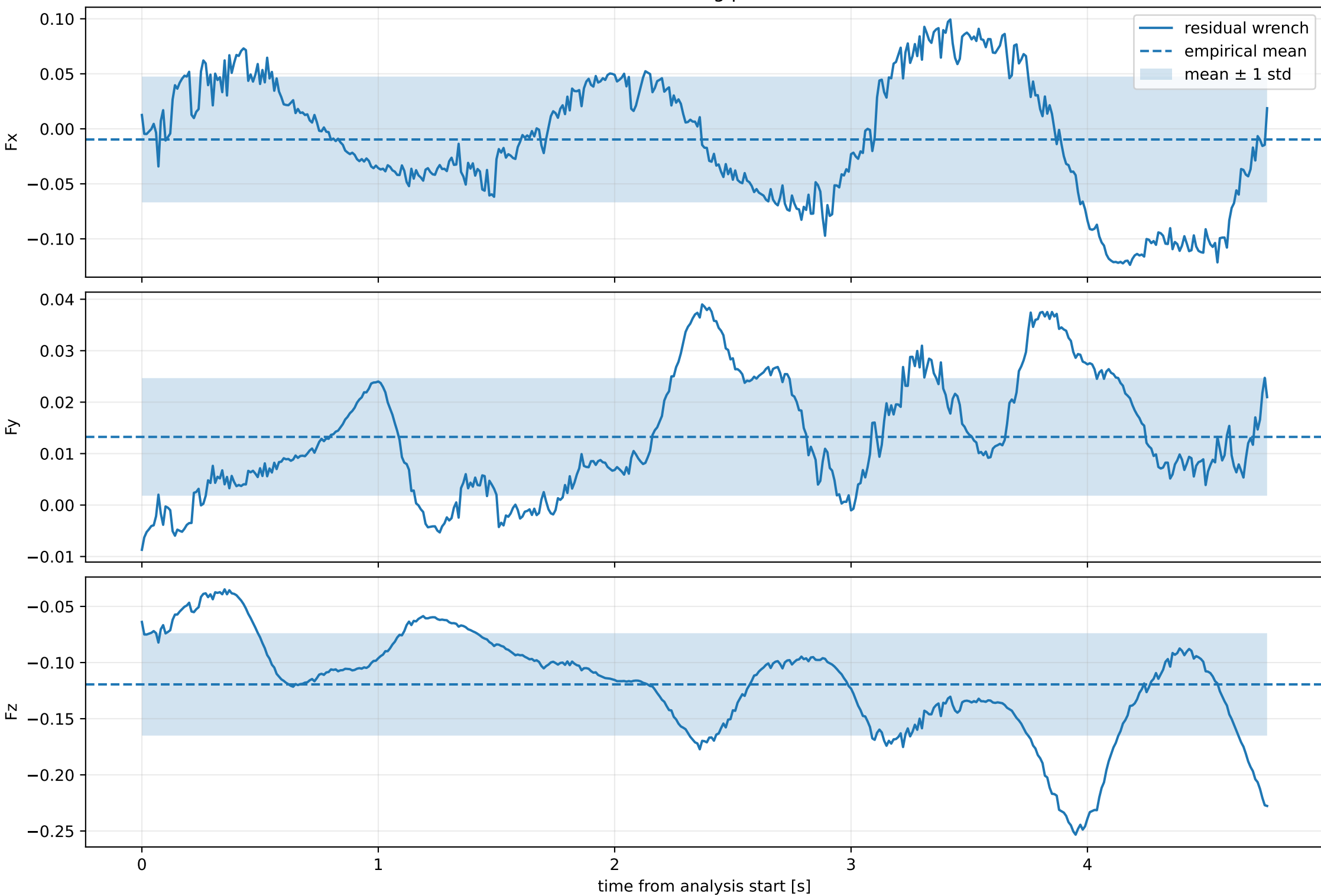
Parameter-only open-loop pose: roll / pitch / yaw



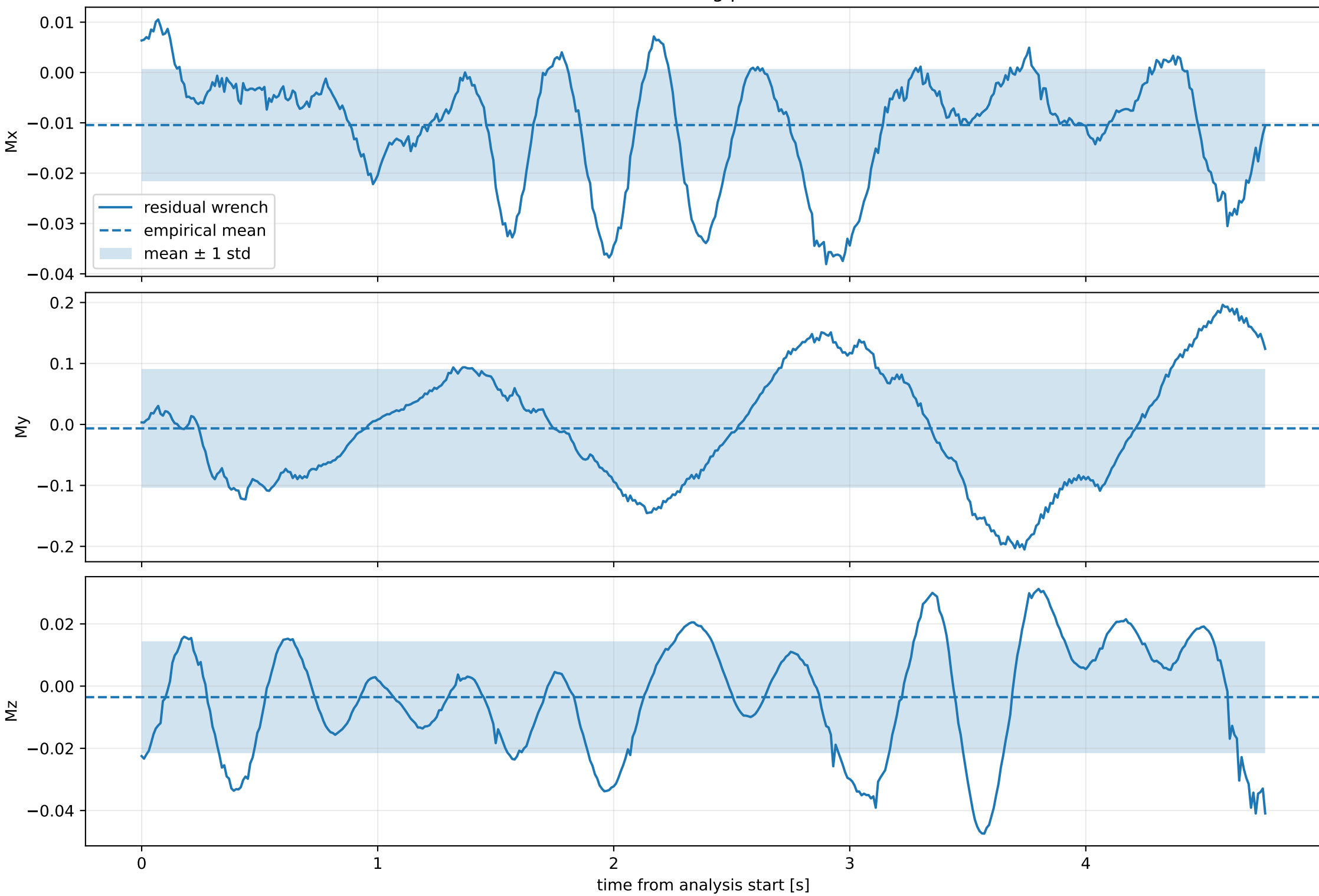
Parameter-only open-loop rollout (separate diagnostic)



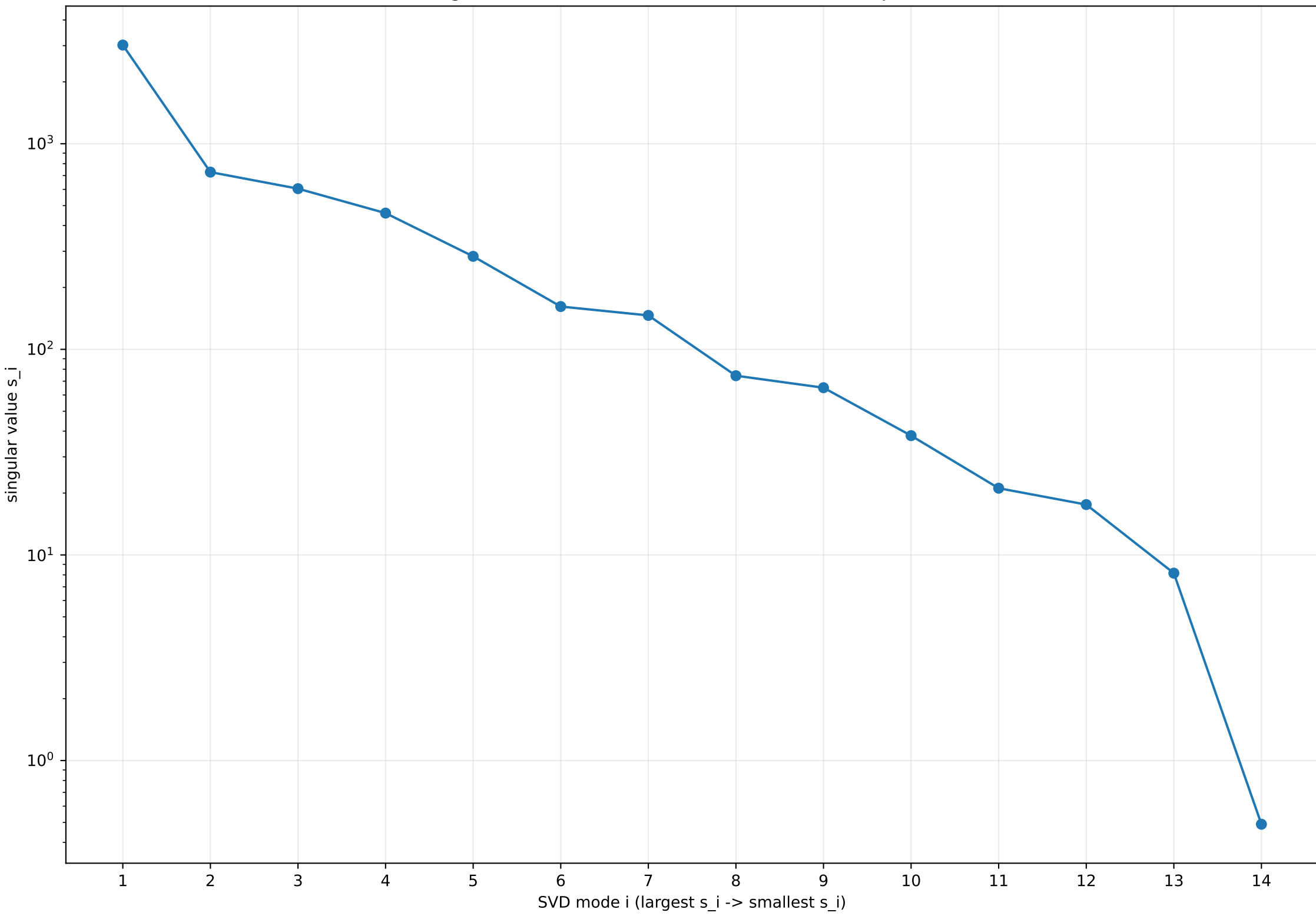
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



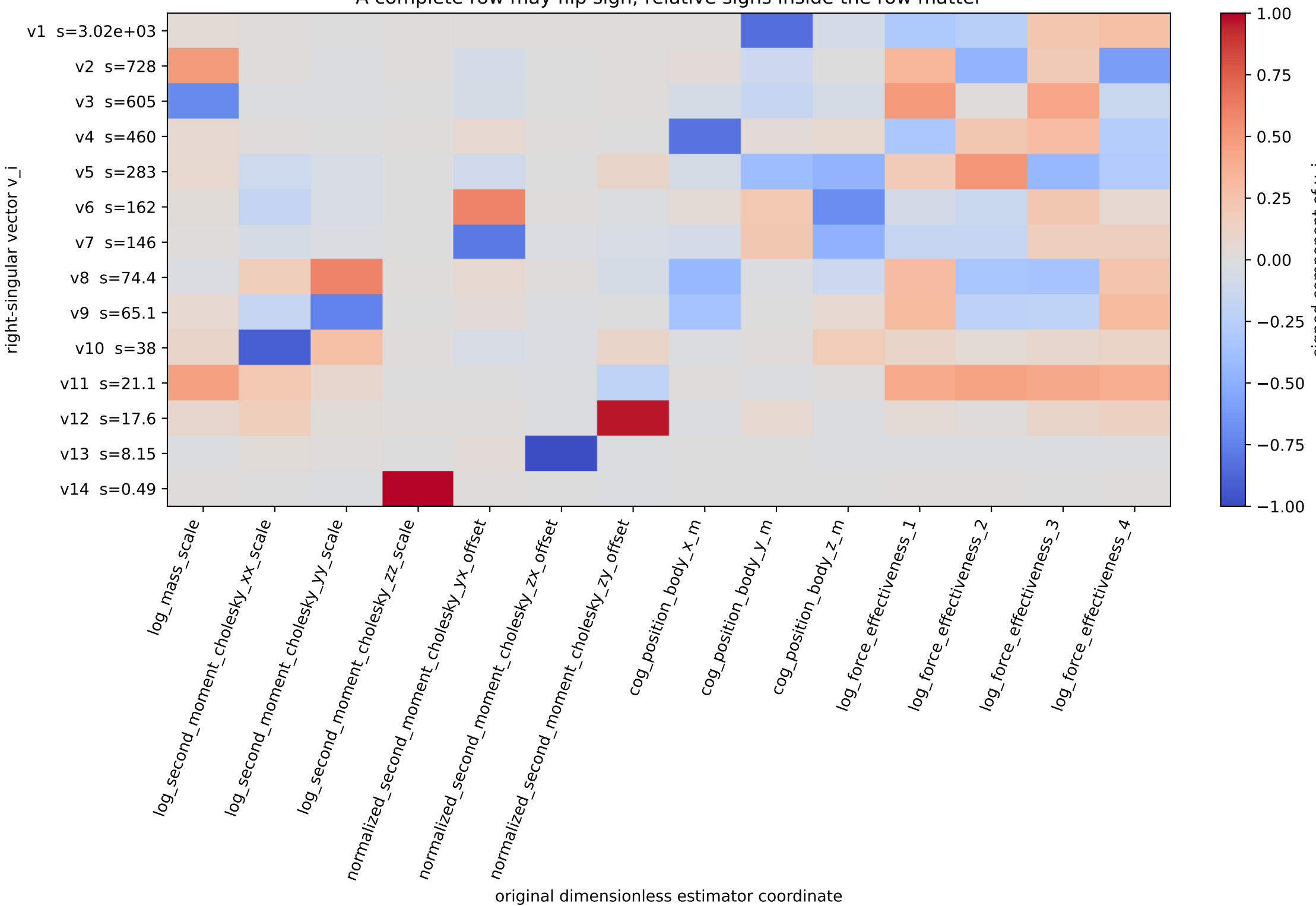
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36585438283

inertia J [kg m²]

[0.065007659 -3.776793e-05 1.9772909e-05]
[-3.776793e-05 0.065512976 2.1854979e-05]
[1.9772909e-05 2.1854979e-05 0.12899286]

principal moments [kg m²]

[0.065004845, 0.065515776, 0.12899287]

CoG position in body frame [m]

[-0.002532024, 0.0008484322, 0.009921401]

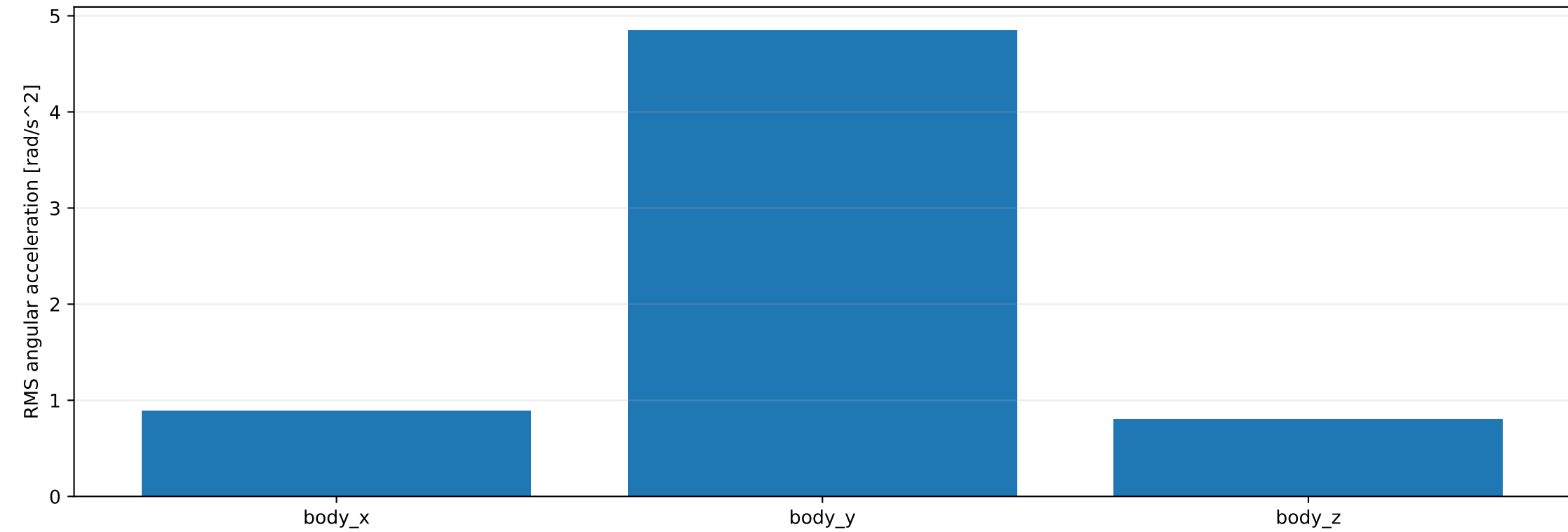
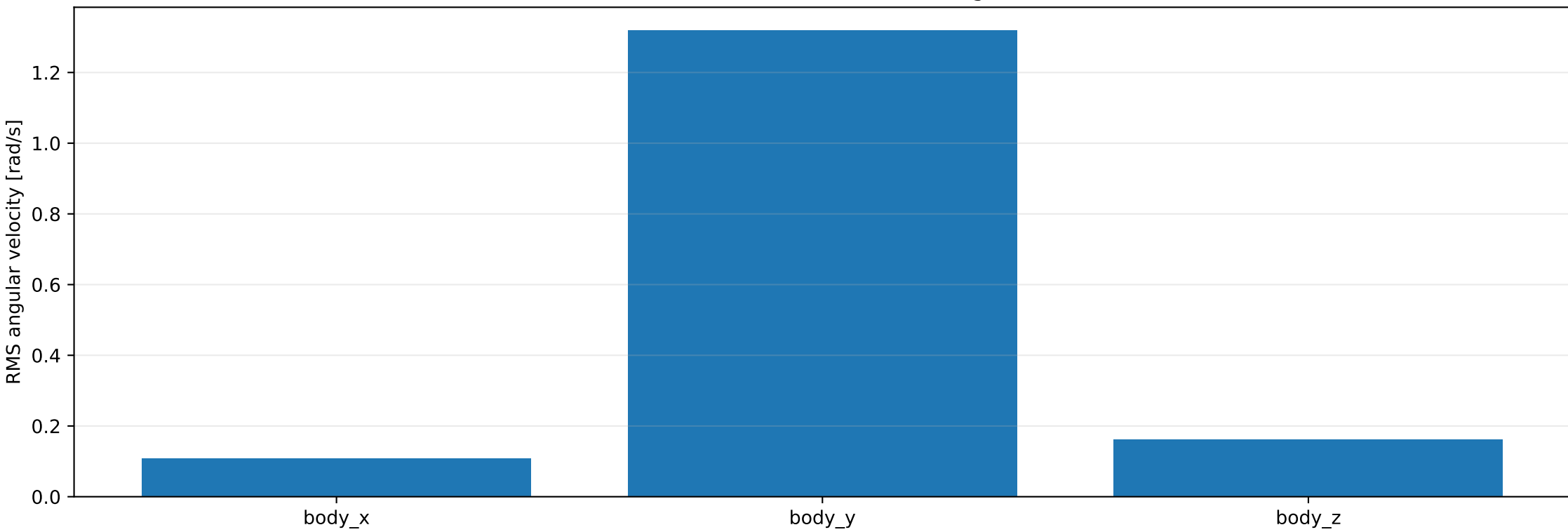
rotor force effectiveness

[0.96219288, 0.97217679, 0.95726357, 0.95221586]

selected delay [s]

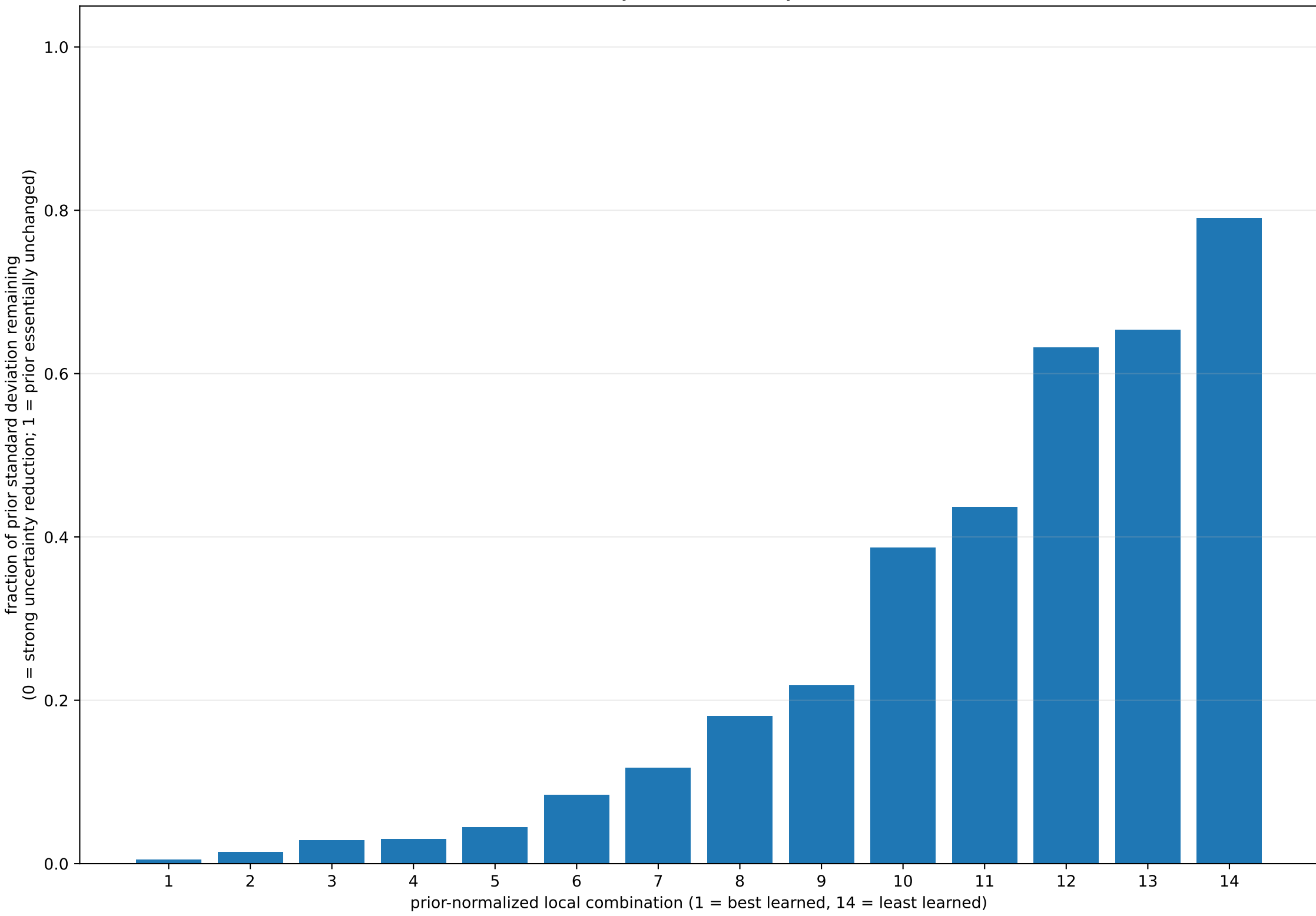
0.0652617932

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

